Statistical analysis of mass spectrometry generated peptidomics data

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Peptidomics is the comprehensive characterization of endogenous peptides from a biological sample mainly by HPLC and mass spectrometry. This analytical methodology allows the detection of a multitude of single peptides, even in complex mixtures. Label-free quantitative peptidomics has gained popularity in recent years as it has fewer sample processing steps, no limitation on sample number, and no need for pricy isotope labeling, and thus is well suited for large-scale studies (1). In a general label-free quantitative peptidomics experiment, peak intensity information for each peptide is compared across multiple LC-MS runs, where run-to-run variability introduces a number of difficulties in data analysis and needs to be carefully adjusted for (Fig.1). In addition, a higher percentage of missing values are often present in multi-run LC-MS peptidomics data due to a number of factors, including the inability of the mass spectrometer to select and fragment every peptide in every run.

Here, we outline the data processing and statistical analysis steps using R software and Bioconductor packages for label-free quantitative peptidomics experiments, with an emphasis on setting up criteria for missing value evaluation and addressing the run-to-run variability issue, which can lead to several major problems in label-free experiments (Fig.2).

Missing values in MS experiments are unavoidable and problematic to downstream data analysis. A criterion should be set up against which the presence/absence of a feature will be judged. Popular criterion is the “modified 80% rule”, where a feature is kept if it contains non-zero values in over 80% of samples in any group (2). Features that passed will proceed to missing value imputation, followed by batch correction where peak intensities will be corrected for batches and potentially other confounding factors. The statTarget R package utilize QC runs and injection order for signal correction (3). Incorporate (pooled) QC runs is highly recommended in large-scale label-free MS experiments, as QC runs can be used to adjust for signal drift within and across batches and calculate coefficient of variation (CV) of each feature, which serves as a filter to remove features with poor repeatability across samples (example injection order is shown in Fig.3).

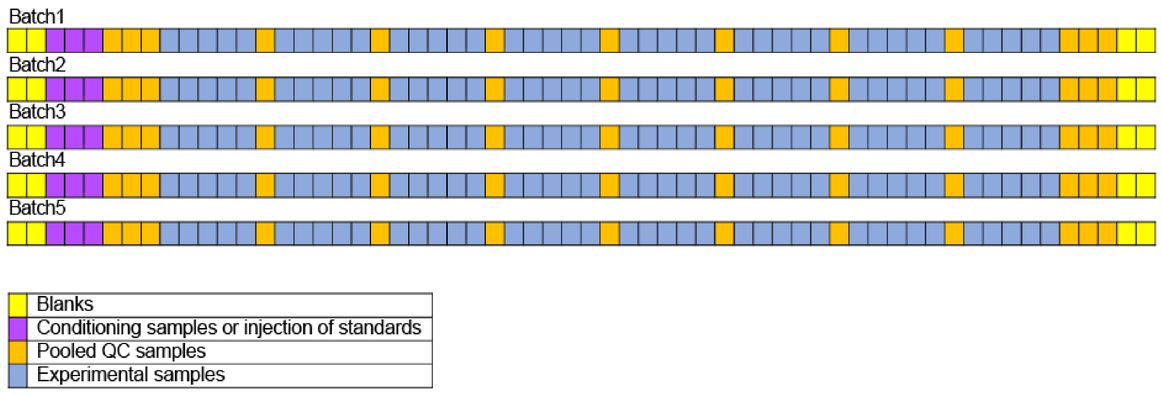

Figure 3. Example injection order for label-free quantitative peptidomics data acquisition. Sample LC-MS injection order should be randomized and includes periodic pooled QC injections throughput the entire data acquisition phase of an experiment. Figure adapted from Trpp, B. A., et al. (4).

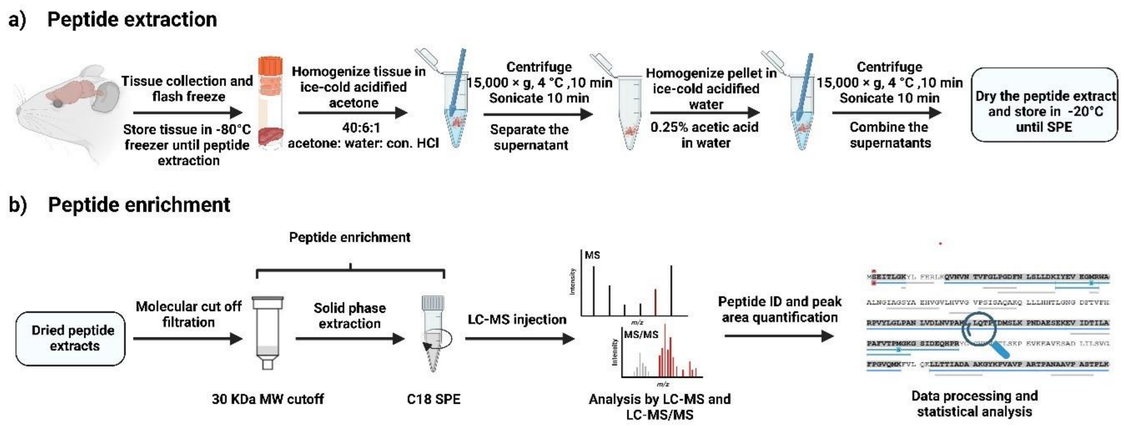

Figure 1. General workflow for label-free quantitative peptidomics. (A) Peptide extraction from tissue collection to peptide enrichment. (B) Peptide enrichment of peptide extracts followed by LC-MS analysis, peptide identification and peak area quantification.

After batch correction, MS peak intensity is usually heavily skewed and not normally distributed. Log transformation is a recommended first step for data processing. Normalization is a data processing step that usually follows log transformation, as it helps remove systemic biases from the data. Scaling aims to adjust for the differences in feature dispersion. However, scaling changes the structure of data more profoundly, and therefore should be performed with greater caution. After data processing, selection of statistical analysis method should be depending on the data distribution. Multiple testing correction should be incorporated after statistical tests due to high-dimensional nature of MS data.

Overall, our analysis pipeline provides researchers with a roadmap for the statistical analysis of their own label-free quantitative peptidomics data generated from a wide variety of different biological sources.

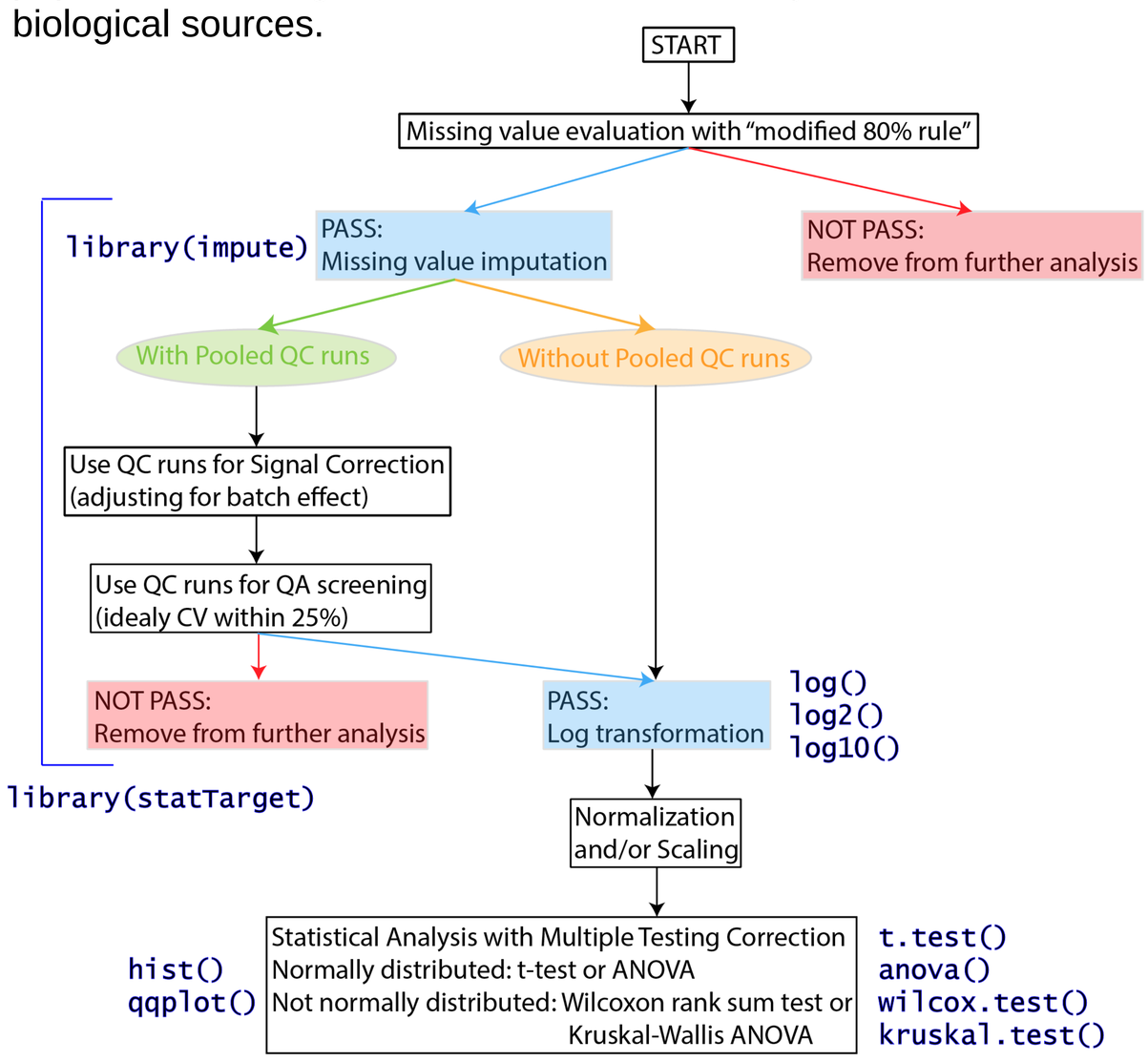

Figure 2. Diagram workflow for label-free quantitative peptidomics data analysis after LC-MS acquisition and feature ID transfer.

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